CHEATSHEET · QUICK REFERENCE

Lab 4 — Shared Storage: NAS (NFS / CIFS) and SAN (iSCSI)

Export file-level NFS and CIFS shares, then build a block-level iSCSI target and connect an initiator so the LUN appears as a plain disk.

NFS export (file-level NAS)

```
echo "/srv/nfs/share 127.0.0.1(rw,sync,no_subtree_check)" >> /etc/exports
```
Declare the export

```
exportfs -ra
```
Re-export all shares

```
showmount -e 127.0.0.1
```
List exports

```
mount -t nfs 127.0.0.1:/srv/nfs/share /mnt/nas
```
Mount from the client

iSCSI target and initiator (block-level SAN)

```
truncate -s 500M /srv/lun0.img
```
Create the backing store

```
systemctl restart tgt
```
Load the target definition

```
tgtadm --mode target --op show
```
Verify the target / LUN

```
iscsiadm -m discovery -t st -p 127.0.0.1
```
Discover targets

```
iscsiadm -m node --login
```
Log in -> new /dev/sdX appears

```
mkfs.ext4 /dev/sdX && mount /dev/sdX /mnt/san
```
Format the LUN like a local disk

NAS vs. SAN vs. FC/FCoE

Feature	NFS/CIFS (NAS)	iSCSI (SAN-IP)	FC / FCoE
Access type	File	Block (LUN)	Block (LUN)
Transport	TCP/IP	TCP/IP	FC / Ethernet+DCB
Typical use	User shares	DB, VM datastores	High-throughput SAN
HBA/switch	NIC only	NIC or iSCSI HBA	FC HBA + FC switch
Cost	Lowest	Low	Highest